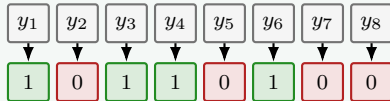


A $\sigma_g > 0$: the rewards differ

one prompt x , drawn from the step's batch B_t

G completions sampled i.i.d.;
together they form one group g



scalar rewards from one shared verifier

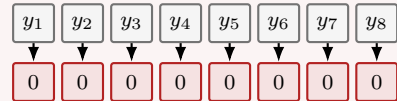
$$\begin{aligned}\mu_g &= \frac{1}{G} \sum_{i=1}^G r_{g,i} = 0.500 \\ \sigma_g &= \sqrt{\frac{1}{G-1} \sum_{i=1}^G (r_{g,i} - \mu_g)^2} = 0.535 \\ A_{g,i} &= (r_{g,i} - \mu_g) / (\sigma_g + \varepsilon) \\ \varepsilon &= 10^{-6}; \text{ unbiased sample variance (ddof = 1)}\end{aligned}$$

Advantage is usable: some $A_{g,i} \neq 0$, so the group carries a learning signal.
This group is counted in $\text{GU} = 1 - \text{ZVF}$, not in ZVF.

B $\sigma_g = 0$: the rewards are all identical

one prompt x , drawn from the step's batch B_t

G completions sampled i.i.d.;
together they form one group g



the same verifier returns the same value for all G

$$\begin{aligned}\mu_g &= \frac{1}{G} \sum_{i=1}^G r_{g,i} = 0.000 \\ \sigma_g &= \sqrt{\frac{1}{G-1} \sum_{i=1}^G (r_{g,i} - \mu_g)^2} = 0.000 \\ A_{g,i} &= (r_{g,i} - \mu_g) / (\sigma_g + \varepsilon) \\ \text{every reward identical} &\Rightarrow \sigma_g = 0\end{aligned}$$

Advantage is identically zero: every $A_{g,i} = 0$, so the group contributes nothing at this step.
It is counted in ZVF; here the cause is all-wrong, but all-correct is identical.

Definitions. $\text{ZVF}_{0,t} = \frac{1}{|B_t|} \sum_{g \in B_t} \mathbf{1}[r_{g,1} = r_{g,2} = \dots = r_{g,G}] =$

$\frac{\text{\# groups whose } G \text{ rewards are identical}}{\text{\# groups in the step's batch } B_t}$

$\text{GU}_{0,t} = 1 - \text{ZVF}_{0,t}$ (gradient utilisation: the share of groups that carry within-group contrast)

Exact ties require zero variance. The report's thresholded ZVF_ε instead counts sample variance $\widehat{\text{Var}} \leq 10^{-6}$, including near-ties. They agree for these binary examples, but differ after micro-jitter.

Measured composition. Qwen3-8B on a held-out GSM8K slice via the Tinker API, $G = 8$, $T = 1.0$, 200 problems $\times$ 3 seeds: mean ZVF = 0.16, hence GU = 0.84.

$$\text{ZVF} = 12.7\% + 3.2\% = 0.16$$



all-correct
12.7%

mixed — contrast present, so counted in GU: 84.1%